DO YOU KNOW?

Any part of pickle which sticks out of oil gets acted by yeast decreasing the acid level. Once this acid level goes down, the bacteria starts acting on the pickle, making it slimy and slippery. This is how **pickles get spoilt**.

Latest techniques in food preservation

4.2.7 Food Irradiation

Food irradiation is a process of food preservation in which food is exposed to ionizing energy – *radio isotope cobalt 60 and cesium-137*. The electromagnetic *radiation suppresses* the growth of most *microorganisms*.

Hospitalized patients, who have compromised immune systems and *astronauts in space*, consume irradiated foods. More than forty years of scientific research show that this process is safe. The radiant energy kills the bacteria in the food, but it does not touch the food directly.

The uses of food irradiation are:

- *To avoid the use of harmful chemical compounds* in insect disinfestations of stored products and microbial decontamination of spices.
- *To extend the shelf life* of meat, poultry and sea foods by killing microorganisms which cause spoilage.
- *To replace the chemicals* used for slowing sprouting in tubers and bulbs and delay ripening of fruits.

4.2.8 Vacuum Packing

Vacuum packing is a process that removes air from the package prior to sealing. This method involves placing items in *a plastic film package, removing air from inside, and sealing the package*. This technology is *widely used* all over the world in the *packaging of milk and milk products, juice etc.*

Facts!

China is the largest apple producing country in the world.

In banana cultivation, Tamil Nadu tops the other states in India.

SUMMARY

- Millions of fruits and vegetables are produced each year and they are lost due to poor processing and preservation.
- Hence all fresh foods have to be preserved to avoid spoilage.
- Preservation ensures increase in shelf life of foods, the quality, the edibility, retention of nutritive value and retention of original colour of food
- The principles of food preservation techniques are prevention or delay of microbial decomposition, prevention and delay of self-decomposition of food, and prevention of damage caused by mechanical causes, insects and rodents.
- Preservation methods include traditional techniques and modern industrial techniques.
- In traditional techniques curing, freezing, boiling, pickling and salting, etc., helps to preserve the food for future use.
- Advanced techniques like vacuum packing, irradiation improves the quality and shelf life of the food.

Chapter 4_070-087.indd 84 31-03-2018 13:21:41